

Correctness Scores Are Unreliable Safety Certificates for Reasoning-Step Rollbacks:

A Counterfactual Study of Step Removability

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Abstract

Process reward models (PRMs) score the local correctness of reasoning steps, and correctness scores are natural candidates for deciding which steps of a generated trajectory can be pruned or rolled back. We test whether they can *license* such interventions. Our counterfactual protocol—control, deletion, position-and-length-matched *placebo* deletion with its own matched control, and semantic-preserving *paraphrase*—is applied to 600 rating-balanced PRM800K steps (over 23,000 regenerated continuations), with replications on two further generators, a within-checkpoint thinking-mode toggle, and 300 ProcessBench steps. Deletion gains concentrate almost entirely in steps that are both low-rated and semantically harmful (+23.3 pp; difference-in-differences +25.4 pp): erroneous steps act as *negative anchors* for no-thinking generation. An apparent generic “restart” benefit under a shared control disappears once each placebo receives its own cutoff-matched control, and paraphrasing the erroneous step performs like keeping it. Toggling the same generator into extended thinking makes the anchor vanish (+21.8 → −3.0 pp). Turning scores into rollback policies, we certify the net accuracy change per deleted step under deployment weighting with nested train/calibrate/test selection: certificates are split-unstable, and their calibration is checkpoint-dependent—one trained PRM certifies in every split yet meets its budget on only 9 of 20 test splits (realizing −6.3 pp per deletion), while another transfers on most—so a correctness-based certificate cannot be trusted without counterfactual validation. A single probe-and-rollback generation removes all detectable excess harm. Correctness scores nominate rollback candidates; they are unreliable safety certificates. We release the audited dataset, pipeline, and benchmark.

1 Introduction

Process reward models assign scores to individual steps of a reasoning trajectory and have become a standard tool for verifying and steering large language model (LLM) reasoning (Lightman et al., 2024; Uesato et al., 2022; Wang et al., 2024). As long chains of thought grow expensive, systems have begun pruning generated reasoning with attention-based redundancy scores (Choi et al., 2025), step entropy (Li et al., 2025), and likelihood-preserving token deletion (Singh and Hakkani-Tür, 2026). Step-level *correctness* scores are the obvious next candidate—PRMs are exactly the models trained to say which steps are bad—but reusing them as deletion signals embeds a silent assumption: that a step’s correctness score tells us what happens to downstream reasoning if the step is *removed*. Before that reuse hardens into practice, we test the assumption directly.

This assumption conflates distinct properties of a step: it can be locally correct yet contribute nothing later steps use; locally incorrect yet arrive after the trajectory is already doomed; or incorrect in a way that actively *anchors* the continuation to a bad path, so removing it rescues the run. Whether deletion helps is a *counterfactual* property of the step-generator pair, not a syntactic property of the step in isolation.

We study this gap directly. Starting from PRM800K (Lightman et al., 2024), we sample 600 target steps balanced across human ratings (−1/0/+1), annotate each step’s semantic contribution (Essential / Redundant / Harmful), and measure what actually happens when a fixed generator (Qwen3-8B; Qwen Team, 2025) continues the trajectory with and without the step, four runs per condition, 2,400 paired outcomes. Because deleting *any* step also forces the generator to re-plan—which can itself change outcomes—we additionally run a position-and-length-matched *placebo* dele-

tion for 511 of the 600 steps (1,514 runs), letting us decompose raw deletion effects into a target-specific component and a generic restart component. An LLM judge scores final answers, audited against 200 human adjudications (headline sensitivity ≤ 0.13 pp).

Three findings emerge. **First, correctness and contribution are correlated but far from interchangeable** (Cramér’s $V = 0.53$; 34.5% of steps off the expected diagonal), and deletion outcomes track the *interaction*: only steps that are both low-rated and Harmful yield large reliable gains (+23.3 pp), while every other rating or type group is statistically indistinguishable from zero. The pattern replicates on a second no-thinking generator (phi-4: anchor effect +33.7 pp) and on external ProcessBench trajectories from twelve source models; toggling the *same* Qwen3-8B checkpoint into thinking mode collapses the anchor effect (+21.8 $\rightarrow$ -3.0 pp), as does a long-CoT distill—extended thinking itself neutralizes bad prefixes. **Second, the anchor-group gain is semantic through and through.** A single-control placebo suggests a +10.0 pp generic “restart” component, but that contrast is confounded: half the matched cutoffs fall before the target and silently remove it. Giving every placebo cutoff its *own* control (1,514 new runs) shows matched innocuous deletions are inert in every group, and the anchor effect is fully target-specific (+25.4 pp, difference-in-differences); the paraphrase control shows it travels with meaning—rewording the erroneous step performs like keeping it, while deleting it gains +19.2 pp over the paraphrase (a *negative anchor*). **Third, scores nominate but do not certify.** We budget the *certified net accuracy change per deleted step* (a one-sided 95% cluster bound), which repairs a subtle flaw in transition-rate “harm”: independent regeneration makes a null deletion on a stochastic step look 25% harmful. Under deployment weighting with nested train/calibrate/test selection over 20 splits, correctness-based certificates are unstable (even random certifies in 9/20 splits, meeting the budget in only 2), and transfer is checkpoint-dependent: one trained PRM certifies in every split yet meets its 1 pp budget in only 9 of 20 on test (-6.3 pp per deletion realized); another transfers in 16 of 19—and correctness quality does not predict which. The rating-+1 mass that compression actually targets (92% of natural traffic) never certifies at budgets below 5 pp; a trajectory-state predictor is the most

reliable certifier (+8.8 pp per deletion, budget met in 18/20). The information gap, not the intervention, is the bottleneck—and a single probe-and-rollback generation removes all detectable excess harm while raising the net gain to +11.5 pp per candidate.

Our contributions: (i) a conceptual separation of step correctness, semantic contribution, and retrospective removability, with pre-specified estimands; (ii) a reusable counterfactual protocol—control / target deletion / matched placebo / semantic-preserving paraphrase—with an audited automatic evaluator; (iii) an empirical characterization of *negative reasoning anchors*, replicated across two no-thinking generators and an external multi-model benchmark, and shown to vanish when the same generator thinks before answering; (iv) a signal audit spanning dataset ratings, deployed PRM scores, uncertainty, and counterfactual importance; and (v) a causal risk-coverage and rollback benchmark showing that the transfer of correctness-based safety certificates across problems is unreliable and checkpoint-dependent, while cheap counterfactual validation removes all detectable excess harm. We release all data, code, and audits.

2 Related Work

Process supervision and PRMs. Step-level supervision improves credit assignment over outcome-only rewards (Uesato et al., 2022; Lightman et al., 2024; Cobbe et al., 2021). Math-Shepherd derives step labels from rollout success probabilities (Wang et al., 2024); Setlur et al. (2025) argue for rewarding *progress* (prospective advantage) rather than local correctness; generative PRMs verify steps with explicit reasoning (Khalifa et al., 2025). Benchmarks probe what PRMs detect: ProcessBench locates first errors (Zheng et al., 2025), and PRMBench separates fine-grained error dimensions including redundancy (Song et al., 2025). None measures what happens when a step is *removed*—the counterfactual layer our protocol supplies.

Reasoning compression and pruning. Token- and step-pruning methods compress chains of thought using attention-based redundancy (Choi et al., 2025), step entropy (Li et al., 2025), or likelihood-preserving greedy deletion (Singh and Hakkani-Tür, 2026); none uses PRM correctness scores as its deletion signal, which is precisely the reuse whose safety we quantify before it becomes

practice. These works optimize for preserving the answer of an already-correct trajectory; we additionally quantify the *repair* regime and show the two regimes respond to different signals.

Deletion as intervention. Ablation-style causal analyses of reasoning typically mask content at the token level or measure answer-probability shifts. Our design instead runs full *regenerated* continuations—the deployment-relevant intervention—and controls the regeneration with matched placebos that carry their *own* controls, which proves essential: a shared-control placebo manufactures an apparent +10 pp “restart” component that disappears, and understates the semantic effect, once each cutoff is read against its own control (Section 6).

3 Constructs and Estimands

We distinguish four properties of a step s_t in trajectory h : (1) *local correctness*, the PRM800K rating in $\{-1, 0, 1\}$; (2) *semantic contribution*, whether the content of s_t is needed by (Essential), irrelevant to (Redundant), or misleading for (Harmful) downstream reasoning; (3) *prospective advantage*, the change in success probability from *taking* the step (Setlur et al., 2025); (4) *retrospective removability*, the change in success probability from *deleting* the step after the fact and regenerating. This paper measures (4) and its relation to (1) and (2).

For step i , let $Y_{ij}^c, Y_{ij}^t \in \{0, 1\}$ be final-answer correctness under **Control** (step kept) and **Target deletion**, with $K=4$ independent runs per condition; matched steps (511/600) additionally receive one to four **Placebo** runs Y_{ij}^p (1,514 in total), applying the same truncation operator at a different, length-matched step p ; every placebo run further receives an *own control* C_p (prefix kept through p ; 1,514 runs), so the placebo can be read against its own cutoff rather than the target’s, $\Delta_i^{\text{sem,DiD}} = \Delta_i^{\text{tgt}} - (\bar{Y}_i^p - \bar{Y}_i^{C_p})$ (Section 6). Pre-specified estimands (fixed in the released analysis plan before the extension cohorts ran; we did not use an external registry):

$$\Delta_i^{\text{tgt}} = \bar{Y}_i^t - \bar{Y}_i^c, \quad \Delta_i^{\text{plc}} = \bar{Y}_i^p - \bar{Y}_i^c, \quad \Delta_i^{\text{sem}} = \Delta_i^{\text{tgt}} - \Delta_i^{\text{plc}} \quad (1)$$

plus run-level *discordance* events *danger* ($Y^c=1, Y^t=0$ on independent draws) and *benefit* ($Y^c=0, Y^t=1$). Runs are independent fresh-session samples with no shared randomness across conditions; pairing control run j with deletion run

j is a fixed bookkeeping convention whose expected discordance rate equals $p_i^c(1 - p_i^t)$ under any pairing, so danger/benefit rates estimate those products—independent-sample discordance, never within-seed counterfactual flips (Section 9 builds the causal safety metric). Δ^{sem} is a *placebo-corrected* contrast; the paraphrase control probes its semantic reading directly (Section 6). All confidence intervals are 95% percentile intervals from 5,000-replicate cluster bootstraps over *target steps*; re-clustering by problem leaves headline intervals essentially unchanged (Table 8). Fixed seeds are released.

4 Data and Intervention Protocol

Step cohort. From PRM800K phase-2 test trajectories (Lightman et al., 2024) over MATH problems (Hendrycks et al., 2021) we sample 600 target steps, 200 per rating, balanced over early/middle/late positions (201/201/198). Semantic contribution was annotated twice, with both passes completed before any deletion outcome existed: an initial AI-assisted pass (165 Essential / 201 Redundant / 234 Harmful) and a human-calibrated re-labeling pass (198 / 199 / 203); the passes disagree on 164/600 steps (27.3%), an explicit reliability bound; a blinded four-annotator pilot quantifies human agreement on the taxonomy (Appendix B). The frozen statistical pipeline behind Tables 1–2 uses the *initial* pass as its primary label set ($-1 \times$ Harmful anchor: 178 steps); the calibrated pass (anchor: 160 steps, 154 shared) drives the correctness-vs-contribution crosstab below, extension-cohort sampling, and a robustness re-estimation of the anchor effects (Table 8; the headline survives either pass). Under the calibrated labels, rating and type are associated (Cramér’s $V = 0.528$) with 65.5% of steps on the expected diagonal—correlated, but with a third of the cohort off-diagonal (Figure 1, appendix).

Interventions. Both interventions are prefix truncations of the same trajectory: the **Control** prompt contains the problem plus steps 1: t , the **Target-deletion** prompt contains steps 1: $t-1$ —operationally a *rollback* of the most recent visible step, not an interior splice—and Qwen3-8B (temperature 0.7, top- p 0.8) regenerates the continuation, four fresh runs per condition. The **Placebo** condition applies the identical truncation at a different step p of the same trajectory (before or after t ; 51%/49% in practice) whose token length is within $\pm 20\%$ of the target’s, choosing the closest available

position; 511/600 steps admit such a match (1,514 placebo runs), and the 89 unmatched steps differ from matched ones on no outcome-relevant covariate (max position SMD 0.298). The placebo-cutoff steps are benign: none carries rating -1 (Table 9).

Outcome evaluation and audit. An LLM judge (Qwen3-8B, temperature 0; prompts released) marks final-answer correctness against ground truth. We audited 200 stratified outputs with human adjudication: agreement 93.5% (F1 0.931), pair-transition agreement 91.7%, and a maximum per-condition bias of 2.3 pp—below the pre-set gates (90% / 5 pp). Substituting all 200 adjudicated labels into the frozen tables moves the overall target effect by -0.13 pp and the placebo-corrected effect by -0.05 pp—observed sensitivity on the audited subset, not a bound on the remainder; no conclusion moves under substitution (Table 5). Because the judge shares a checkpoint with the primary generator, a deterministic PRM800K-style normalization+SymPy grader independently regrades all 6,314 primary runs: 93.0% agreement, headline effects within 0.9 pp (Appendix A).

Replication and mechanism cohorts. Four extensions reuse the identical prompt protocol. (i) *Cross-generator*: 300 steps (all 100 available rating $-1 \times$ Harmful anchors, 100 rating-0, 100 rating-1; 103 stably-correct; positions balanced) re-run with **phi-4** (14B), 3 runs per condition, reusing the frozen matched-placebo indices; the identical 2,691-task list is additionally rerun with **DeepSeek-R1-Distill-Qwen-14B** (long-CoT, served locally; sampling and judge unchanged) and—holding the checkpoint fixed—with **Qwen3-8B’s own thinking mode enabled** (same provider, prompts, and sampling as the primary runs).¹ (ii) *Semantic-preserving paraphrase* (condition C3): for a 240-step strong-control subset (80 anchors / 80 stably-correct / 80 rating-0/1 fragile), an independent model (Gemini 2.5 Flash) paraphrases the target step preserving its exact meaning *including errors*; 221/240 paraphrases passed automatic validity checks (7.9% excluded) and each ran 4 continuations, alongside 4 fresh matched-placebo runs (C2). (iii) *Deployed PRM scoring*: five PRMs score all 600 steps—three trained checkpoints (Qwen2.5-

¹R1-Distill was unavailable via the API provider at freeze time; phi-4 matched the 7–14B no-thinking budget (all 2,691 continuations parsed). The local long-CoT rerun parsed 99.0%; its 28 unparseable continuations, balanced across conditions, score as incorrect.

Group	Steps	Δ^{tgt} (pp)	95% CI
Overall	600	+7.21	[+3.88, +10.58]
rating = -1	200	+22.00	[+15.41, +28.42]
rating = 0	200	+0.25	[−5.41, +5.90]
rating = $+1$	200	−0.63	[−5.54, +4.21]
Essential	165	−1.21	[−7.24, +4.78]
Redundant	201	+1.12	[−3.87, +6.06]
Harmful	234	+18.38	[+12.28, +24.44]
$-1 \times$ Harmful	178	+23.31	[+16.57, +30.24]

Table 1: Target-deletion effects on final-answer accuracy (600 steps, 2,400 paired runs, cluster bootstrap over steps; initial-pass semantic labels—the calibrated-pass anchor estimate is $+20.9$ pp, Table 8).

Math-PRM-7B, full-trajectory token-level head; Math-Shepherd’s Mistral-7B PRM; RLHFlow’s Llama3.1-8B PRM trained on DeepSeek rollouts) and two prompted LLM-judge PRMs (Gemini 2.5 Flash, phi-4). (iv) *External validation*: 300 ProcessBench steps (Zheng et al., 2025) spanning GSM8K, MATH, OlympiadBench, and Omni-MATH trajectories from twelve source models (Section 7).

5 Full-Cohort Deletion Results

Deleting target steps helps on average ($+7.21$ pp, Table 1), but the average is misleading: the entire reliable gain sits in the rating = -1 column and the Harmful row, peaking at $+23.31$ pp for their intersection. Correct steps (rating $+1$) and Essential steps show near-zero *average* effects—yet not because deletion is safe there. Danger discordance is substantial everywhere: on 15.1% of paired draws the control sample is correct and the deletion sample wrong (an estimate of $p^c(1 - p^t)$, not an individual causal flip; Section 3), and even within rating = -1 this rate is 14.9%. The near-zero group means arise from offsetting discordance in both directions, not from inertness.

Trajectory state dominates raw effects. Stratifying by control stability makes the state-dependence explicit (Figure 2, appendix): steps whose four control runs are all wrong gain $+33.4$ pp from deletion (nothing to lose), while steps whose control runs are all correct *lose* 12.1 pp (CI $[-15.5, -8.9]$). A deployment policy never observes these counterfactual states, which is exactly why raw deletion gains overstate what a policy can capture (Section 9).

Step-level stability. With four paired runs per step, 377/600 steps never change outcome; 99

are strongly beneficial (gain in ≥ 3 runs) and 57 strongly harmful. Of the 178 low-rated Harmful steps, 29% are strongly beneficial to delete— $2.5\times$ the rate elsewhere (11%), but far from a guarantee.

Cross-generator replication. The pattern is not a Qwen3-8B artifact. Rerunning 300 steps with phi-4 (3 runs per condition) reproduces every qualitative conclusion: overall target effect +12.4 pp, anchor-group target effect +33.7 pp with own-controlled placebos inert (−1.3 pp) and a DiD semantic component of +34.9 pp (CI [+25.8, +44.7]), and rating-0/rating-1 target effects indistinguishable from zero (+1.7 and +2.0 pp). Effect *magnitudes* are generator-dependent, and step-level effect signs agree for 65% of the 80 steps with nonzero effects under both generators—well above chance but far from deterministic.

Long-CoT replication. The same 2,691 tasks under DeepSeek-R1-Distill-Qwen-14B—which thinks before answering—change the picture qualitatively. Anchor-step control accuracy rises to 77.7% (the model recovers from erroneous prefixes unaided), the overall deletion effect shrinks to +3.4 pp, and the anchor-group effect of +7.0 pp is not about the anchor at all: deleting a matched *innocuous* step helps just as much (+8.0 pp against its own control), and the DiD semantic component is −1.0 pp (CI [−7.8, +6.3]). Only 82/300 steps shift outcome at all: negative anchoring—and with it most of deletion’s value—vanishes here, leaving only a small content-agnostic truncation benefit.

Thinking-mode contrast. The R1 replication changes model, scale, and training data at once; to isolate the mode, we rerun the same tasks on the *same* Qwen3-8B checkpoint with thinking enabled (identical provider, prompts, sampling; thinking activates on every task, mean 2,514 reasoning tokens; 12.2% of runs yield no parseable answer and are scored incorrect, balanced across conditions). On the anchor cell, whose no-thinking control accuracy is 26.8% with a +21.8 pp deletion effect, thinking lifts control accuracy to 83.3% and the deletion effect drops to −3.0 pp (CI [−8.0, +2.3]; DiD semantic −3.5 pp, n.s.); only 77/300 steps shift outcome at all, and step-level effect signs agree with the no-thinking runs at chance (56%, $n=32$). The collapse of negative anchoring is attributable to extended thinking itself, not to the R1 checkpoint.

Group	Δ^{tgt}	$\Delta_{\text{vs C}}^{\text{plc}}$	$\Delta_{\text{own}}^{\text{plc}}$	$\Delta_{\text{DiD}}^{\text{sem}}$ (CI)
Overall (511)	+7.97	−0.57	+0.46	+7.52 [+2.8, +12.4]
rating = −1 (169)	+22.78	+9.47	−0.69	+23.47 [+14.4, +32.5]
rating = 0 (176)	+0.99	−5.63	+2.84	−1.85 [−9.7, +6.0]
rating = +1 (166)	+0.30	−5.42	−0.90	+1.20 [−5.9, +8.4]
Harmful (201)	+18.41	+3.57	−1.74	+20.15 [+12.0, +28.4]
−1×Harmful (151)	+23.84	+9.99	−1.60	+25.44 [+16.1, +34.9]

Table 2: Placebo decomposition (pp; initial-pass labels). $\Delta_{\text{vs C}}^{\text{plc}}$: placebo vs. the *target’s* control (frozen contrast); $\Delta_{\text{own}}^{\text{plc}}$: vs. its *own* control C_p (1,514 new runs); $\Delta_{\text{DiD}}^{\text{sem}} = \Delta^{\text{tgt}} - \Delta_{\text{own}}^{\text{plc}}$. Matched innocuous deletions are inert everywhere.

6 Mechanism: Anchors, not Restarts

Does deletion help because the *content* of the step was harmful, or because *any* deletion lets the generator restart and resample? Table 2 answers with two placebo contrasts. Against the target’s control, placebo truncation inside the low-rated groups looks strongly positive (+9.99 pp)—the reading a single-control design would report as a “restart benefit”. But this contrast confounds removing step p with moving the cutoff: 51% of matched cutoffs fall *before* the target, so those placebo prefixes silently drop the erroneous step too. Splitting by side is diagnostic: cutoffs before the target yield +21.1 pp against the shared control (they remove the anchor), cutoffs after it yield −6.8 pp (they keep it). Giving each placebo cutoff its *own* control—new C_p runs that keep step p and truncate after it—removes the confound: $\Delta_{\text{own}}^{\text{plc}}$ is null in every group (−1.6 pp on anchors, +0.5 overall), and the difference-in-differences semantic effect *rises* to +25.4 pp (CI [+16.1, +34.9]) on anchors—essentially the entire raw effect. Deleting matched innocuous content does nothing; what matters is whether the *anchor* leaves the prefix. (An identical-prompt subset bounds generation-batch drift at +3.1 pp, within noise; Table 12.)

Two methodological corollaries. First, placebo deletions need their own controls: matching position and length is not enough—the single-control contrast *understated* the anchor’s semantic component by ~ 12 pp and manufactured a spurious restart component. Second, the rating-0/+1 “semantic” effects of the single-control analysis (+5.7 to +6.6 pp) vanish under the difference-in-differences (−1.9 to +1.2 pp, n.s.): target-specific removability lives *only* in the anchor group.

Paraphrase triangulation. The placebo isolates *that* the target matters; a semantic-preserving paraphrase isolates *why* (Table 3). In the anchor group,

Contrast (pp)	Anchor	Stable-corr.	Fragile-neutral
C1–C0 (raw deletion)	+20.3	−11.9	+24.1
C1–C2 (vs. placebo)	+15.6	+16.8	+0.0
C3–C0 (paraphrase)	+1.7	−8.3	+20.9
C1–C3 (vs. paraphrase)	+19.2	−2.7	+4.1
C2–C _p (placebo, own ctrl.)	−0.7	−0.8	+3.9

Table 3: Four-condition strong controls (240 steps $\times$ 4 runs; 80 per group; paraphrase available for 221). Anchors: paraphrase $\approx$ control while deletion beats both—the *content* is the anchor. Fragile-neutral: deletion, placebo, and paraphrase all match the uplift over the fragility-conditioned control (regression to the mean; the own-control row shows the operator is inert). Stably-correct: even meaning-preserving rewording hurts (−8.3 pp).

replacing the erroneous step with an independent-model rewording that preserves its (wrong) meaning leaves accuracy at control level (+1.7 pp, CI [−6.2, +9.9]), while deleting it gains +19.2 pp over the paraphrase (CI [+8.2, +30.1]): the harm travels with the *semantics*, not the surface form—exactly the signature the negative-anchor account predicts. The fragile-neutral group shows the mirror image: a large raw gain (+24.1 pp) that placebo deletion—and even paraphrase—matches (C1–C2 = 0.0 pp), i.e. nothing target-specific; the own-control row attributes the shared uplift to conditioning on fragile control runs rather than to any benefit of the operator. And stably-correct steps are harmed even by faithful rewording (−8.3 pp, CI [−14.7, −3.0]), showing that correct reasoning chains are sensitive to perturbations that preserve meaning—deletion is not the only risky edit. A blinded 140-item human fidelity audit backs the C3 premise (97.1% faithful or minimally deviating; zero of 46 audited anchor-group paraphrases corrected the original error), and the 19 excluded paraphrases resemble the 221 retained (Table 14).

Eight fully human-verified cases (78 audited continuations) illustrate the mechanism space and ship with full continuations.

7 External Validation on ProcessBench

Does any of this survive outside PRM800K trajectories and outside human step ratings? We sample 300 ProcessBench steps (Zheng et al., 2025)—152 human-annotated *first-error* steps and 148 *locally-correct* steps, balanced across GSM8K, MATH, OlympiadBench, and Omni-MATH (75 each) and drawn from trajectories written by twelve different source models—and run the identical protocol (3 runs per condition; 204/300 steps admit a length-

matched placebo; gold answers matched for 99.5% of records).

The core dissociation transfers. Deleting first-error steps yields +24.8 pp (CI [+16.9, +32.7]); on the placebo-matched subset, the difference-in-differences semantic component is +35.4 pp (CI [+22.8, +48.9]) with own-controlled placebos slightly *negative* (−8.4 pp)—erroneous steps in other models’ reasoning are negative anchors too. Deleting locally-correct steps yields −2.0 pp raw (CI [−8.3, +4.1]) and a −7.1 pp (n.s.) semantic component: not usefully removable. One boundary emerges: the DiD semantic component is significant on GSM8K (+37.3 pp) and MATH (+21.1 pp) but attenuates on OlympiadBench (+6.0 pp, CI crosses zero) and vanishes on Omni-MATH (−0.6 pp)—and own-controlled placebos show no restart benefit there either (+6.3 pp, n.s.): on problems near the generator’s competence ceiling, nothing target-specific survives. Negative anchors are a phenomenon of *recoverable* trajectories.

8 Do Correctness Signals Predict Removability?

If PRM-style signals suffice for pruning, they should predict the two events a policy cares about: *danger* (destroying a correct run) and *benefit* (rescuing a wrong one). We evaluate step-level signals with 5-fold cross-validation grouped by problem (no problem appears in both train and test), L2-regularized logistic models, and cluster-bootstrap CIs (Table 6, appendix).

The asymmetry matters. Signals rank *benefit* discordance moderately well: low rating and Harmful type genuinely enrich for rescuable runs. But for *danger* discordance—the safety-relevant direction—the rating alone is anti-informative (AU-ROC 0.422): among runs whose control draw is correct, low-rated steps are *not* safer to delete, and conditioning on the rating actively misleads. Only features that peek at trajectory state (control stability; effectively extra compute) move danger prediction appreciably. Cheap local correctness is calibrated for “was this step wrong?”, not for “will removing it break the run?”. These AU-ROC/AUPRC numbers are exploratory diagnostics on independent-sample discordance; the safety conclusions of Section 9 rest on held-out net effects and certified bounds, not on these rates.

Deployed PRMs, uncertainty, and counterfactual signals. Because “PRM scores” in deployment are continuous model outputs rather than dataset ratings, we scored the identical 600 steps with three trained PRM checkpoints (Qwen2.5-Math-PRM-7B, Math-Shepherd, RLHFlow-DeepSeek) and two prompted LLM-judge PRMs, plus step entropy/NLL under teacher forcing and a masked answer-probability drop $I_{\text{mask}} = \log p(y^* | h) - \log p(y^* | h \setminus s_t)$ (Table 7, appendix). The best trained PRM tracks the human rating almost exactly (danger AUROC 0.57 vs. 0.56), and the other two trained checkpoints sit at or below chance (0.47, 0.51)—so the construct gap is not an artifact of auditing labels instead of scores. Token-level uncertainty carries no removability information at all. Two signals stand out modestly: cross-PRM disagreement (benefit AUPRC 0.34 across the five, close behind the best single checkpoint’s 0.37), supporting uncertainty-based *abstention*, and the counterfactual mask drop—informative but gold-answer-dependent, hence offline-only.

9 From Scores to Policies: Risk–Coverage

We convert the audit into a deployment question: if a system deletes the top- k steps ranked by a signal, does it *certifiably* not lose accuracy, and how many individual steps does it hurt? A tempting metric—the rate at which control-correct runs come back wrong after deletion—is not causal harm: outcomes are independent samples, so a null deletion on a step with $p^c=p^t=0.5$ would show 25% “harm” from sampling disagreement alone. We therefore budget the *certified net accuracy change per deleted step*: a policy is certified at budget b if the one-sided 95% lower confidence bound (problem-cluster bootstrap) of the mean Δ^{tgt} over its selected steps stays above $-b$. Alongside, we report step-level *tail risk*: the share of selected steps with observed $\hat{\Delta}_i < 0$, minus the exact share expected under the null $p^t=p^c$ (Binomial(4, $\hat{p}$) per step). Policies: random; correctness rankings (rating-first, Harmful-first, intersection-first); the three trained PRM scores; the model-D/E predictors of Section 8; an in-sample oracle.

On the *balanced* cohort this certification is degenerate: the case–control mix is so error-rich that the mean deletion effect is +7.2 pp (LCB +4.4), and every policy—including random—certifies deleting *all 600 steps* at every budget. Certifica-

tion only becomes meaningful in the deployment frame, weighting steps to the natural chosen-path prevalence (next paragraph): there, deleting all traffic has LCB -4.4 pp and certifies nothing below a 5 pp budget.

Balanced cohort vs. natural prevalence. The cohort is case–control balanced (200 steps per rating); deployment traffic is not. Our reference frame is the annotator-chosen trajectory path of the phase-2 test split—the steps a deployed sampler would actually carry—comprising 16,481 rated steps: 31 rated -1 , 1,224 rated 0, 15,226 rated $+1$ (Table 11). Chosen paths are error-scarce (annotators route around bad candidates); the rated-but-rejected alternatives form a far larger error pool (26,256 rated completions, 23.2% rated -1), and our rating- -1 targets are drawn from that pool and spliced onto the chosen prefix (198/200; the other rating tiers are chosen-path steps). Reweighting to the chosen-path prevalence therefore assumes spliced alternatives behave like the rare naturally-carried -1 steps, and adjusts rating composition only (the cohort is position-balanced by design; a joint rating $\times$ position reweighting gives -0.3 pp, CI $[-5.0, +4.5]$, indistinguishable from rating-only). Under these weights the average deletion effect moves from +7.2 pp to -0.5 pp (CI $[-5.1, +4.0]$), with danger and benefit discordance near parity (9.8% vs. 9.3%; Table 10): in natural traffic, indiscriminate deletion has no upside while the danger does not shrink. We present natural-weight numbers as a reference-frame *sensitivity analysis* resting on this explicit transport assumption; balanced-cohort numbers are diagnostic contrasts. Neither is a deployment estimate.

What certifies in natural traffic. Under natural weights with in-sample selection (Table 13, appendix), correctness rankings certify only the error tiers at a 1 pp budget: rating-first covers 2.4% of traffic (+9.8 pp per deletion), Harmful-first 3.8% (+17.0), and the best trained PRM score 0.02% (+51). These deletions are *error correction*, not compression—the rating- $+1$ mass that a pruning system actually wants to remove (92% of traffic) never certifies at any budget below 5 pp. The learned predictors certify 66–68% of traffic in-sample (+4.2 to +4.5 per deletion); the oracle certifies 96%.

Honest nested selection. Certification must survive selection. Table 4 repeats the exercise with

Policy	Certifies	Met	Coverage	Net/del. (pp)	Tail exc.
Random	9/20	2/9	0.00	−3.7	0.043
rating-first	20/20	15/20	0.49	+0.9	0.044
Harmful-first	12/20	6/12	0.20	−1.8	0.064
PRM (Qwen2.5-Math)	20/20	9/20	0.18	−6.3	0.089
PRM (Math-Shep.)	19/20	16/19	0.33	+6.4	0.000
PRM (RLHFlow)	14/20	8/14	0.05	+3.3	0.038
Predictor (D)	15/20	13/15	0.13	+20.8	0.000
Predictor (E)	20/20	18/20	0.58	+8.8	0.010

Table 4: Nested policy evaluation, natural prevalence, 1 pp budget, 20 outer 50/50 problem splits. Predictors are trained on the calibration half only, with inner problem-grouped 5-fold cross-fitting so every calibration-side score is out-of-fold; coverage is chosen on those calibration scores (cluster-robust LCB) and evaluated exactly once on the untouched test half. “Certifies”: splits with nonzero certified coverage; “Met”: certifying splits whose realized test net effect stays within the 1 pp budget; “Coverage”/“Net/del.”/“Tail exc.”: medians over splits.

nested selection: predictors are trained on the calibration half only with inner cross-fitting (all calibration-side scores out-of-fold), coverage is chosen on calibration, and test is scored exactly once. Three failures emerge. *Instability*: certified coverage swings from 0 to 0.9 across splits for most policies—with ~ 150 calibration problems, a 95% bound is noisy enough that even *random* certifies in 9 of 20 splits, meeting the budget in only 2. *Checkpoint-dependent transfer*: the trained PRM with the best danger diagnostic (Qwen2.5-Math, AUROC 0.57) certifies in every split yet meets the 1 pp budget in only 9 of 20, realizing **−6.3** pp per deletion with the largest tail excess (0.089); Math-Shepherd, by contrast, transfers in 16 of its 19 certifying splits (+6.4 pp). Two PRMs trained for the same construct yield opposite safety behavior—a certificate’s validity depends on the checkpoint, and correctness quality does not predict which. *Weak payoff*: rating-first meets the budget in 15 of 20 splits but realizes only +0.9 pp per deletion at half-cohort coverage. The state-aware predictor E is the most reliable certifier (18/20 met, +8.8 pp, tail excess 0.010). A certificate obtained on one sample of problems is not a license on the next unless the signal carries removability information—and which checkpoints do cannot be known in advance.

Rollback: one probe run buys most of the safety. Finally, we simulate the cheapest counterfactual validator: generate *one* probe continuation after a candidate deletion, roll back unless the probe answers correctly, and evaluate on held-out runs. Applied to all 600 steps, this raises the

net gain per candidate from +7.2 pp (LCB +4.4) to +11.5 pp (LCB +9.3) and cuts the observed harmed-step share from 12.2% (stochastic floor 11.0%) to 3.0%—*below* its 6.8% floor, i.e. no detectable excess harm survives—at a cost of one extra generation per candidate (63% acceptance). A do-nothing control-split shows the floor is calibrated (4.7% observed vs. 3.7% expected). Two caveats keep this honest: the probe is scored by the same gold-answer judge—an oracle relative to deployment, so the simulation upper-bounds a one-probe validator and presupposes one—and the gain is not mere extra sampling, since matched truncations that *keep* the target are inert against their own controls ($\Delta_{\text{own}}^{\text{plc}}$ +0.5 pp overall, Table 2). Validation is not merely necessary; here it is cheap relative to the value it protects.

10 Discussion and Conclusion

Our results do not indict PRMs at their trained task: low ratings do flag wrong steps, and wrong steps are where deletion *benefit* concentrates. The failure is direction-specific—scores carry almost no information about deletion *danger*, which is governed by trajectory state and generator recoverability. Mechanistically, erroneous content behaves as an attractor: continuations conditioned on the anchor commit to its error. The oracle–policy gap quantifies the headroom for better signals; the obvious candidates do not close it (Section 8), and the informative ones cost extra computation or gold answers. The constructive path is architectural: the nominate–probe–rollback loop already recovers most of the oracle’s safety at bounded cost.

Across 600 audited PRM800K steps, four generator configurations, and over 23,000 regenerated continuations, step correctness and step removability dissociate: deletion gains concentrate in low-rated *Harmful* steps; the apparent restart component under a shared control disappears once each placebo receives its own cutoff-matched control, leaving a gain that travels with the step’s erroneous content; and correctness signals—dataset rating, deployed PRM scores, uncertainty—nominate rollback candidates but certify safety unreliably: their calibration is checkpoint-dependent and transfers unpredictably. Systems that delete reasoning content should nominate with scores, validate counterfactually, and budget harm explicitly—and PRM benchmarks should test whether scores license such interventions.

Limitations

Generator scope. Primary results use Qwen3-8B without thinking; replications add phi-4, DeepSeek-R1-Distill-Qwen-14B, and a within-checkpoint thinking-mode toggle (300 steps, 3 runs each). The two no-thinking models agree qualitatively but not step-wise (anchor DiD semantics: +25.4 vs. +34.9 pp; sign agreement 65%); thinking mode on the same checkpoint neutralizes anchors, and the R1 distill replicates the collapse—so anchor-driven gains should not be assumed beyond no-thinking generation. All generators are mid-size open models on mathematical reasoning; we make no claims about frontier models or other domains. The R1-Distill replication failed at freeze time via the API provider (Section 4) and was completed later on local hardware; phi-4 remains the frozen second generator. **Case-control cohort.** Rating-balanced sampling (200 per rating) supports subgroup contrasts, not deployment averages: reweighted to the natural prevalence of PRM800K phase-2 test steps (0.2% rated -1), the mean deletion effect is -0.5 pp and anchor steps are rare. Every coverage number is a fraction of the balanced cohort (Table 10). **PRM coverage.** The deployed-PRM audit covers three trained checkpoints—Qwen2.5-Math-PRM-7B, the original Math-Shepherd PRM (Wang et al., 2024), and an RLHFlow Llama3.1-8B PRM trained on DeepSeek rollouts—plus two *prompted* LLM-judge PRMs; trained *generative* PRMs (Khalifa et al., 2025) remain uncovered. **Semantic label reliability.** Step types come from AI-assisted annotation with human calibration; the two label passes disagree on 27.3% of steps, and the blinded annotation pilot (Appendix B) shows that trained humans agree only moderately on the semantic taxonomy, with disagreement concentrated on the essential/redundant boundary. Type-based subgroup boundaries are therefore approximate; the pilot covers 120 of 600 steps, and no adjudicated relabeling of the full cohort exists. The headline interaction survives under either label pass, but finer type-level claims inherit this label noise. **Judge noise and self-evaluation.** The 93.5% agreement and ≤ 0.13 pp substitution sensitivity are measured on 200 audited Qwen3-8B outputs and do not bound error on the unaudited remainder; phi-4 and ProcessBench continuations are scored by the same judge pipeline without a fresh per-generator human audit. The deterministic-grader regrade (Section 4; Appendix A) bounds self-evaluation bias on the pri-

mary cohort only—the extension cohorts have no deterministic regrade. **Placebo scope.** Matched placebos exist for 511/600 primary steps (position $SMD \leq 0.298$); the external cohort’s placebo matching uses character length ($\pm 20\%$) rather than tokenizer length, and 96/300 external steps admit no match. **Offline policy evaluation.** Risk-coverage and rollback numbers reuse frozen runs; an online system would face distribution shift from its own deletions. The rollback probe is drawn from the same run pool it is evaluated against, which can flatter acceptance quality slightly. In-sample coverage selection flatters every policy; the nested train/calibrate/test protocol of Section 9 is the honest reading. With $K=4$ runs per condition, per-step harm attribution is noise-limited: the tail-risk excess over the exact stochastic floor is a conservative signal of truly harmed steps, not a per-step identification. Rollback presupposes a verifier and spends one probe generation per candidate, which must be netted against pruning savings in any deployment accounting. **Paraphrase validity.** C3 paraphrases passed automatic meaning/length checks (7.9% excluded), and a blinded 140-item human fidelity audit rated 97.1% faithful or minimally deviating, with zero meaning changes among the 46 audited anchor-group paraphrases; two neutral-group paraphrases (4.3% of that group’s audited items) were judged meaning-changed. C3 conclusions rest on this audited premise.

Ethical Considerations

The study uses the public PRM800K dataset (MIT-licensed problems from MATH) and open-weight models; no personal data is processed. Beyond the authors, four annotators and one adjudicator recruited by the authors labeled reasoning steps and paraphrase fidelity for public math problems (about one hour each); they were informed of the study’s purpose, and the annotation sheets collected no personal information. Deleting reasoning steps is an accuracy/efficiency intervention; our safety framing (harm budgets) is about answer correctness, not user harm. We release artifacts to support reproduction and auditing.

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A Judge Audit Detail

Audited outputs	200
Binary agreement	93.5%
Correct-class F1	0.931
Pair-transition agreement (60 pairs)	91.7%
Max condition bias	2.3 pp
Δ overall target effect after substitution	−0.13 pp
Δ placebo-corrected effect after substitution	−0.05 pp

Table 5: Judge audit summary. Gates (agreement $\geq$ 90%, bias $\leq$ 5 pp) pass; all headline numbers are insensitive to full substitution of adjudicated labels.

B Robustness and Disclosure Tables

Annotation pilot. A blinded pilot re-annotated 120 cohort steps: two independent human labels per step from four annotators in two pairs, 61.7% raw agreement ($\kappa = 0.40$), with disagreement concentrated on the essential/redundant boundary. Where annotators agreed, their consensus tracked the frozen outcomes: 93% of consensus-Harmful steps carry rating -1 , and the placebo-corrected deletion effect on consensus-Harmful steps is $+22.9$ pp. A third annotator adjudicated all 46 disagreements, yielding final labels for all 120 pilot steps with 82% adjudication accuracy.

PRM rating $\times$ human-calibrated contribution type

Cramer’s $V=0.528$, expected-diagonal share=65.5% (full cohort: $N=600$ steps)

	Essential	Redundant	Harmful
rating -1	25 4.2% of all steps	15 2.5% of all steps	160 26.7% of all steps
rating 0	55 9.2% of all steps	115 19.2% of all steps	30 5.0% of all steps
rating 1	118 19.7% of all steps	69 11.5% of all steps	13 2.2% of all steps

Outlined cells form the expected rating–contribution diagonal; 34.5% are off-diagonal.

Figure 1: PRM800K rating vs. human-calibrated semantic contribution for the 600-step cohort. The association is strong (Cramér’s $V = 0.528$) but 34.5% of steps are off-diagonal: local correctness does not determine contribution.

Model (features)	Danger AUPRC	Benefit AUPRC
Base rate	0.151	0.360
A: rating only	0.139	0.432
B: type only	0.208	0.407
C: rating + type	0.140	0.380
D: static context	0.160	0.412
E: + trajectory state	0.263	0.478

Table 6: Out-of-fold prediction of run-level danger (base rate 15.1%) and benefit (base rate 36.0%). The rating is *below chance* for danger (AUROC 0.42, CI [0.36, 0.49]); only trajectory-state features (model E, requiring extra runs) achieve a meaningful margin (danger AUPRC +0.10 over D).

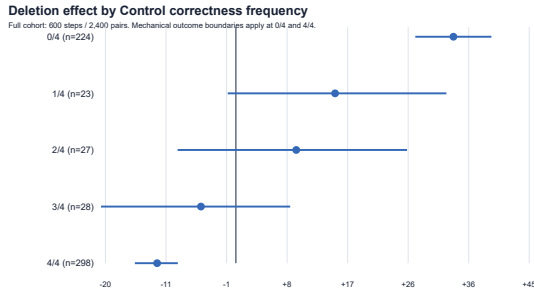


Figure 2: Deletion effect by control-run stability. Gains exist only where control already fails; stably-correct trajectories are harmed (−12.1 pp).

Predictor features. Model D uses 12 static features: indicators rating= 0 and rating= +1; indicators type=Redundant and type=Harmful (initial pass); their four interactions; position-bin indicators (middle, late); $\log(1 + \text{target tokens})$; $\log(1 + \text{prefix tokens})$. Model E adds the control-correct frequency (13 features). Both are L2-regularized logistic models trained with 5-fold cross-validation grouped by problem; every policy score is an out-of-fold prediction.

Judge configuration. The judge is Qwen3-8B served at temperature 0, prompted with the problem, the gold answer, and the generated final answer, and instructed to return a JSON verdict on mathematical equivalence; outputs with no extractable final answer are failed by rule. The audit sampled 200 outputs stratified by condition and verdict; two annotators labeled blind to condition, with disagreements adjudicated. Full prompts, seeds, and adjudication records ship with the release.

Deterministic grader. Because the LLM judge and the primary generator share a checkpoint, we regrade all 6,314 primary runs with a deterministic grader: PRM800K-style answer normalization

Signal (step-level)	Danger		Benefit	
	AUROC	AUPRC	AUROC	AUPRC
Base rate	—	0.15	—	0.24
PRM800K rating	0.56	0.17	0.61	0.34
Qwen2.5-Math-PRM-7B	0.57	0.18	0.61	0.37
Math-Shepherd PRM (7B)	0.47	0.14	0.54	0.26
RLHFlow PRM (8B)	0.51	0.15	0.59	0.28
LLM-judge PRM (Gemini)	0.53	0.16	0.51	0.25
LLM-judge PRM (phi-4)	0.49	0.14	0.48	0.23
PRM disagreement (5 models)	0.53	0.16	0.62	0.34
Step entropy	0.46	0.15	0.48	0.24
Step NLL	0.50	0.15	0.49	0.22
Masked answer-prob drop	0.56	0.19	0.57	0.30

Table 7: Step-level signals vs. deletion outcomes on the identical 600-step cohort (danger/benefit = any correct-to-wrong / wrong-to-correct transition; signals oriented so higher warns against / flags for deletion). The best benefit AUPRCs reach 0.34–0.37 over a 0.24 base—a real lift—while the best danger AUPRC (0.19) barely exceeds its 0.15 base.

followed by a SymPy equivalence check. It agrees with the LLM judge on 93.0% of runs, and substituting its verdicts reproduces the overall and anchor deletion effects within 0.9 pp, bounding self-evaluation bias on the primary cohort; the full comparison ships with the audit artifacts.

Estimate	Cluster	<i>n</i>	Effect (pp)	95% CI
Anchor Δ^{tgt} , initial	step	178	+23.31	[+16.6, +29.9]
Anchor Δ^{tgt} , initial	problem	178	+23.31	[+16.2, +30.5]
Anchor Δ^{sem} (legacy), initial	step	151	+13.85	[+7.0, +20.8]
Anchor Δ^{tgt} , calibrated	step	160	+20.94	[+13.8, +28.1]
Anchor Δ^{tgt} , calibrated	problem	160	+20.94	[+13.5, +28.4]
Anchor Δ^{sem} (legacy), calibrated	step	137	+14.84	[+7.5, +22.1]
Overall Δ^{tgt}	step	600	+7.21	[+3.9, +10.5]
Overall Δ^{tgt}	problem	600	+7.21	[+3.7, +10.7]

Table 8: Label-set and clustering robustness (fresh 5,000-replicate bootstraps). The anchor effect survives the calibrated label pass and problem-level clustering. Rows marked *legacy* use the shared-control placebo contrast of Eq. 1, superseded by the own-control difference-in-differences of Table 2 (+25.4 pp); they are retained for comparability with the frozen analysis plan.

Rating of placebo-cutoff step	Runs	Share
+1	900	59.4%
0	167	11.0%
unrated	447	29.5%
−1	0	0.0%

Table 9: PRM800K ratings of the 1,514 placebo cutoff steps: the placebo never truncates at a known-bad step.

C Reproducibility

Primary numbers derive from two frozen master tables (600 steps; 6,314 runs) via a dependency-

	Balanced cohort	Natural prevalence
Overall Δ^{tgt} (pp)	+7.21 [+3.9, +10.7]	-0.52 [-5.1, +4.0]
Run danger rate	8.5%	9.8%
Run benefit rate	15.7%	9.3%
Dangerous-step share	15.2%	17.5%
Beneficial-step share	23.5%	17.6%

Table 10: Balanced-cohort vs. natural-prevalence estimates (weights 0.2%/7.4%/92.4% for ratings -1/0/+1, from 16,481 rated phase-2 test steps).

Frame	n_{-1}	n_0	n_{+1}	Total
Phase-2 test, chosen path	31	1,224	15,226	16,481
Phase-2 test, all completions	6,080	2,002	18,174	26,256
Cohort targets from chosen path	2	198	200	400
Cohort targets from alternatives	198	2	0	200

Table 11: Sampling-frame audit (integer counts). The 600 cohort targets are unique steps over 303 problems; every cohort problem is in the phase-2 test split. Rating-1 targets come almost entirely from rated-but-rejected alternative completions spliced onto the chosen prefix; reweighting (Table 10) uses the chosen-path frame.

free Python pipeline with fixed seeds (cluster bootstrap seed 20260722, downstream 20260723); a single command regenerates every table, figure source, and the numeric manifest consumed by this paper. The extension cohorts add 17,126 runs (cross-generator 2,691; long-CoT replication 2,691; thinking-mode toggle 3,582; strong controls 1,843 = 884 paraphrase C3 + 959 placebo C2, one of the 960 scheduled C2 generations having failed at the provider; ProcessBench 2,412; placebo own-controls 3,907 across four cohorts), each with its own checkpointed generation, judging, and analysis script under the corresponding work-stream directory. Robustness artifacts (label-set re-estimation, placebo-cutoff composition, natural-prevalence reweighting, and the calibration/test policy split) are regenerated by dedicated scripts in the same release. The release includes the intervention builder, matched-placebo selector, judge-audit tooling, policy risk-coverage and rollback code, the 120-step blinded annotation pilot, PRM-scoring adapters, the M5 signal-extraction script, and the StepRem benchmark split with its evaluation script.

Cohort / check	$\Delta_{\text{own}}^{\text{plc}}$ (pp)	$\Delta_{\text{DiD}}^{\text{sem}}$ anch. (pp)
Primary, anchors (151)	-1.6 [-6.5, +3.2]	+25.4 [+16.1, +34.9]
phi-4, anchors (100)	-1.3 [-6.6, +3.8]	+34.9 [+25.8, +44.7]
M4 C2, all groups (240)	+0.8 [-2.7, +4.4]	—
Anchor cutoffs before / after	-5.2 / +4.3	—
Batch drift (128 identical prompts)	new - old = +3.1 pp (n.s.)	

Table 12: Own-control placebo (DiD) robustness. Matched innocuous deletions are inert in every cohort and group; the drift row compares fresh C_p runs against frozen runs with byte-identical prompts, bounding generation-batch drift well below the contrasts of interest.

Policy	Budget 1 pp		Budget 3 pp	
	Traffic	Net/del. (LCB)	Traffic	Net/del. (LCB)
Random	0	—	0	—
rating-first	2.4%	+9.8 (+2.0)	12.2%	+1.4 (-1.9)
Harmful-first	3.8%	+17.0 (+1.2)	3.8%	+17.0 (+1.2)
PRM (Qwen2.5-Math)	0.02%	+51.3 (+35.5)	16.3%	+8.2 (-1.8)
PRM (Math-Shep.)	21.8%	+9.6 (+3.0)	31.3%	+5.2 (-2.8)
PRM (RLHFlow)	40.4%	+5.0 (-0.6)	47.7%	+3.1 (-2.4)
Predictor (D)	66.1%	+4.2 (-0.8)	88.9%	+1.1 (-2.9)
Predictor (E)	68.5%	+4.5 (-0.1)	79.7%	+1.5 (-2.7)
Oracle	96.4%	+3.2 (-0.2)	96.4%	+3.2 (-0.2)

Table 13: In-sample causal certification under natural prevalence: largest traffic coverage whose one-sided 95% cluster LCB of net effect per deletion clears the budget, with the realized net (LCB in parentheses). In-sample k -selection upper-bounds every policy; the nested evaluation (Table 4) is the honest reading.

	n	Anchor	Stable	Neutral	Rating -1
Retained	221	33.0%	33.9%	33.0%	34.8%
Excluded	19	36.8%	26.3%	36.8%	36.8%

Table 14: Composition of retained vs. automatically excluded C3 paraphrases (mean target length: 38.0 vs. 32.7 tokens).